

S PRAJWALL NARAYANA

AI Engineer

ACADEMIC CURRICULUM VITAE

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RESEARCH INTERESTS

Convolutional architectures and empirical hyperparameter study; evaluation methodology for machine-learning systems that carry a safety obligation (labelled harnesses, deterministic guardrails, regression gates in continuous integration); inference under real constraints, meaning quantized and offline deployment on low-power hardware for multilingual, low-connectivity settings; and retrieval-augmented and multi-agent LLM systems. I am applying for graduate study (MSc) in artificial intelligence, machine learning and machine-learning systems.

EDUCATION

Dec 2021 – Aug 2025
Bengaluru, India

Bachelor of Engineering (B.E.), Computer Science and Engineering

R V College of Engineering, Visvesvaraya Technological University (VTU)

CGPA 6.66 / 10 (VTU 10-point scale, 10 highest).

- Coursework in machine learning, deep learning, computer vision, distributed systems, and data structures and algorithms.
- Final-year research project: Arogya-Sathi, a multilingual offline healthcare assistant with a measured safety-evaluation harness. Described in full under Research Experience.

TODO: add secondary schooling (institution, board, year, grade) only if a specific application form asks for it. Not in the factual base.

PUBLICATIONS

Three peer-reviewed papers published, two manuscripts under review. TODO: add the full author list and author position for each paper, in the order printed on the published version, plus the DOI for the Springer chapter. Admissions committees read author position. None of this is in the factual base and none was guessed.

- EduConnect: A Smart Solution for Streamlined School Administrations.** *ICT for Intelligent Systems (ICTIS 2025)*, Vol. 9, Springer. Presented in Bangkok, April 2025. A school-administration platform combining video conferencing, secure messaging, interactive content and personalised learning dashboards, aimed at low-resource rural schools. TODO: DOI
- An Empirical Study of ResNet50 Hyperparameter Tuning for Plant Disease Classification.** *IEEE Xplore*, November 2024. A structured sweep of the principal hyperparameters across several datasets, characterising the conditions under which ResNet50 stays reliable for plant-disease detection outside benchmark conditions. ieeexplore.ieee.org/document/10816835
- Web-Server Controlled Rover with Robotic Arm and Object Detection.** *IEEE Xplore*, November 2024. A modular LAN-controlled rover integrating an ESP32, YOLOv3 object detection, a four-degree-of-freedom robotic arm and a live camera feed. ieeexplore.ieee.org/document/10816816

Two further manuscripts are under review. Titles and venues are available on request; they are deliberately not printed here while review is in progress.

RESEARCH EXPERIENCE

Jan 2025 – Jun 2025
R V College of Engineering

Arogya-Sathi: evaluation methodology for a multilingual offline clinical assistant

Final-year research project, student lead

- Research question: how to make the safety behaviour of an LLM-based health assistant measurable rather than assumed, and how far that behaviour survives when the whole stack is quantized down to a single-board computer.
- Method: a deterministic, rule-based layer for crisis detection, emergency triage and drug-interaction screening sits beneath the language model and cannot be overridden by it. Its behaviour is scored by a 62-case labelled evaluation harness that also runs as a merge gate in continuous integration at recall 1.00, so a regression blocks the change rather than reaching a user.
- Results, measured on the labelled set: crisis detection recall 1.00, precision 0.93; emergency detection 1.00 / 1.00; drug screening 1.00 / 1.00. Constructing the harness surfaced a self-harm disclosure the system had genuinely missed, which is the finding that motivated the deterministic layer.
- Systems contribution: the full model stack, including symptom checking, X-ray fracture detection, prescription summarisation and a wellness assistant, runs entirely offline on quantized models on a Raspberry Pi, targeting rural deployment without connectivity. A LangGraph pipeline turns an uploaded report into a plain-language explanation in the patient's own language with a self-verification loop, routing requests across local models.
- Third place at the college-level final-year project showcase.

PyTorch, LangGraph, Ollama, YOLOv8, quantized inference, Raspberry Pi | github.com/Developer1010x/Arogya-Sathi

Oct 2024 – Dec 2024
Remote

Detection of manipulated media (deepfake video and synthetic speech)

RVCE Centre of Excellence · Research Intern, AI and Deep Learning

- Implemented a manipulated-face video detector from first principles, using no pre-built detection APIs, so that the architecture and the failure modes were open to inspection.
- Trained and evaluated on the FaceForensics++ and DFDC benchmarks.
- Extended the same line of work to audio, building a detector for AI-synthesised and cloned speech.

PyTorch, TensorFlow, convolutional networks, computer vision, FaceForensics++, DFDC

2024

Empirical hyperparameter study, ResNet50 for plant-disease classification

Study underlying publication [2]

- Designed and ran a structured hyperparameter sweep across multiple datasets, reporting which settings hold up when the data leaves benchmark conditions rather than reporting a single headline accuracy.

PROFESSIONAL EXPERIENCE

Nov 2025 – present
Bengaluru, India

Colligence Research

Founding team · AI products and platform

AI Engineer (Feb 2026 – present)

- Established the DevOps and SRE function from nothing and now leads it: continuous integration and delivery, infrastructure-as-code with OpenTofu and Ansible, observability, alerting and incident runbooks.
- Building a retrieval-augmented system into the infrastructure layer so that infrastructure-as-code and SRE work (runbook lookup, change context, incident response) is grounded in the organisation's own configuration and history. **In progress.**

Associate AI Engineer (Dec 2025 – Feb 2026)

- Led a small team building the company's first multi-agent LLM pipeline for ERP automation over MCP, keeping long-running multi-step workflows on track without human intervention, and a code-generation service that turns structured input specifications into deployable web applications.

Associate AI Engineer Intern (Nov 2025 – Dec 2025)

- Prototyped the first LLM integrations and agent workflows the production pipeline was later built on.

Dec 2024 – Oct 2025 Bengaluru, India	Multi Tech Managing Intern, Operations and Coordination Coordinated day-to-day operations and a team of four interns: onboarding, task assignment, and reporting between the team and management. TODO: confirm this role before sending. It appears in the house-style CVs but not in the agreed factual base, and it is in none of the five country CVs. If it is real it should be added to the factual base; if not it must be removed from every file that carries it.
Aug 2024 – Sep 2024 Bengaluru, India	Ernst & Young India Associate Consultant Intern, Enterprise Cybersecurity Worked alongside senior consultants on cloud-security and identity and access management engagements; exposure to the TOGAF and IFTAS frameworks.

TEACHING AND ACADEMIC SERVICE

Undergraduate R V College of Engineering	Teaching Facilitator, Skill Lab Supported faculty across five laboratory cohorts in practical coursework and hands-on skill sessions.
Nov 2022 – Aug 2024	National Service Scheme (NSS): Core Team and Content Head Recognised by the RSST Trust for outstanding contribution to NSS activities.
Undergraduate	Knotes Central, student academic-resource platform Co-built as one of a four-person founding team; adopted across 12 or more departments. github.com/Developer1010x/KnotesCentral-Source-Code

AWARDS

- Third place**, college-level final-year project showcase, R V College of Engineering (Arogya-Sathi), 2025.
- Winner**, Department Hackathon, Computer Science and Engineering, R V College of Engineering.
- Recognition by the RSST Trust** for outstanding contribution to NSS activities.

TECHNICAL SKILLS

Machine learning	PyTorch, TensorFlow, convolutional networks, ResNet, YOLOv8, OpenCV, quantization and on-device inference, hyperparameter optimisation, evaluation harness design
LLM systems	Agentic and multi-agent workflows, MCP orchestration, retrieval-augmented generation, LangChain, LangGraph, Ollama, Claude API
Systems and tooling	Python, FastAPI, PostgreSQL, Linux, Docker, CI/CD, OpenTofu (infrastructure-as-code), Ansible, Git, shell scripting

LANGUAGES

English	Language of instruction for the degree, of all three publications, and of daily professional work. TODO: add IELTS or TOEFL score and test date once sat. Most MSc programmes outside India require one, or a formal medium-of-instruction letter from RVCE. No score was supplied and none was invented.
Other languages	TODO: list other languages and levels, or delete this row.

REFEREES

TODO: name two or three academic referees with title, institution, department, email and the relationship (final-year project supervisor at RVCE; supervisor at the RVCE Centre of Excellence; a co-author on one of the published papers). No referee names exist in the factual base, so none are printed and no "references available on request" line is used in their place. Almost every MSc application will ask for these by name.	
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Every red box above marks information that was missing rather than guessed. None may remain when this CV is attached to an application.